

Memorandum

1. The fitted regression equation is

$$\widehat{\text{signed contracts}} = 210.64 + 18.75(\text{Google ads expenditure}),$$

On average, for every additional R1 000 spent on Google Ads, the estimated value of signed installation contracts increases by approximately

$$18.75 \times \text{R1 000} = \text{R18 750}.$$

2. Using $SST = SSR + SSE$, we obtain

$$A = SSR = 311\,977.50 - 126\,437.50 = 185\,540.00.$$

Since $n = 72$,

$$D = n - 1 = 71.$$

3. The mean square error is

$$B = \text{MSE} = \frac{\text{SSE}}{df_{\text{Error}}} = \frac{126\,437.50}{70} = 1\,806.25.$$

The residual standard error is

$$\text{RSE} = \sqrt{\text{MSE}} = \sqrt{1\,806.25} = 42.50.$$

Because the response is measured in thousands of rand,

$$42.50 \times \text{R1 000} = \text{R42 500}.$$

Thus, observed contract values typically deviate from their fitted values by approximately R42 500, as measured by the estimated standard deviation of the residuals.

4. Since the regression degrees of freedom equal 1,

$$\text{MSR} = \frac{\text{SSR}}{1} = 185\,540.00.$$

Therefore,

$$C = \frac{\text{MSR}}{\text{MSE}} = \frac{185\,540.00}{1\,806.25} = 102.72.$$

5. The hypotheses for overall model significance are

$$H_0: \beta_1 = 0 \text{ vs } H_a: \beta_1 \neq 0.$$

(Why is this the same as testing the significance of the β_1 estimate? Overall model significance, with the F-test, compares the null model (with an intercept only, in the simple linear regression case, when $\beta_1 = 0$ or $y = \beta_0 + 0x$) to the full model (in the simple linear regression case, when $y = \beta_0 + \beta_1 x$, assuming $\beta_1 \neq 0$)

The test statistic is $F = 102.72$ (which is the same as $10.14^2 = t\text{-value}^2$ from the β_1 coefficient) with 1 (numerator) degree of freedom and 70 (denominator) degrees of freedom.

The p-value is found to be less than 0.001 (from tables – check that you can calculate it in R with the $pf()$ function), and we reject H_0 at the 5% significance level.

There is sufficient evidence that the model containing Google Ads expenditure provides a statistically significant improvement over the intercept-only model, OR, equivalently, there is evidence of a significant linear association between Google Ads expenditure and the value of signed installation contracts.

6. $R^2 = \frac{SSR}{SST} = \frac{185\,540.00}{311\,977.50} = 0.5947.$

Approximately 59.47% of the observed variation in the value of signed installation contracts is explained by its fitted linear relationship with Google Ads expenditure. The remaining approximately 40.53% is not explained by this simple linear regression model.

7. $r = \sqrt{0.594722} = 0.7712.$

There is a strong positive linear relationship between Google Ads expenditure and the value of signed installation contracts.

8. The Google Ads model provides a better fit based on this criterion because it has a lower residual standard error. Its predictions are associated with smaller typical residual deviations. This comparison is appropriate because both models use the same response variable and the same observations.

9. For Google Ads expenditure of R12 000,

$$x = 12$$

because expenditure is measured in thousands of rand.

$$\hat{y} = 210.64 + 18.75(12) = 210.64 + 225 = 435.64.$$

$$435.64 \times 1\,000 = 435\,640.$$